Electrical and Computer Engineering Course Syllabus

010153103 Algorithms and Data Structures

Semester 1 / Academic Year 2026

Department of Electrical and Computer Engineering (ECE)
Faculty of Engineering, King Mongkut's University of Technology North Bangkok

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1. Course Information

Course Number:	010153103
Course Name:	Algorithms and Data Structures
Credits:	3 (3–0–6)
Contact Hours:	3 hours per week
Course Type:	Required
Prerequisites:	010153002 Computer Programming
Textbook:	Goodrich, M. T., Tamassia, R., & Goldwasser, M. H. *Data Structures and Algorithms in Java*, 6th Edition. John Wiley & Sons, 2014.

2. Course Description

This course requires students to understand basic data structures for computer programming development, including *lists*, *stacks*, *queues*, and *trees*; core algorithmic techniques for *sorting*, *searching*, and *hashing*; and the use of *recursion*. Students will learn to select and apply appropriate data structures and algorithms to solve specific programming problems.

3. Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

1. Demonstrate a good understanding of linear data structures, including arrays, linked lists, stacks, and queues.
2. Implement abstract data types effectively.
3. Understand non-linear data structures, including trees and graphs.
4. Select and use appropriate data structures to solve specific problems.
5. Understand and implement sorting, searching, and hashing algorithms.
6. Choose and apply appropriate algorithms to solve specific problems.

4. ABET Student Outcomes Addressed

ABET Student Outcome (SO)	Course Learning Outcome (CLO)
1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	1–6
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	4, 6
3. An ability to communicate effectively with a range of audiences.	—
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	—
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	—
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	2, 4, 6
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	4, 6

5. CLO Assessment Mapping

Course Learning Outcome (CLO)	SO	Assessment Tools
1. Demonstrate a good understanding of linear data structures, including arrays, linked lists, stacks, and queues.	1	Exams
2. Implement abstract data types effectively.	1, 6	Exams, Assignment
3. Understand non-linear data structures, including trees and graphs.	1	Exams
4. Select and use appropriate data structures to solve specific problems.	1, 2, 6, 7	Exams, Assignment
5. Understand and implement sorting, searching, and hashing algorithms.	1	Exams
6. Choose and apply appropriate algorithms to solve specific problems.	1, 2, 6, 7	Exams, Assignment

6. Weekly Schedule

Week	Topic	Reading
1	Introduction to Algorithms & Data Structures; Programming Review	Ch. 1
2	Java & Object-Oriented Design	Ch. 2
3	Algorithm Analysis (Big-O, Big-Ω, Big-Θ)	Ch. 4
4	Array Lists	Ch. 3
5	Singly Linked Lists	Ch. 3
6	Doubly Linked Lists	Ch. 3
7	Stacks & Queues	Ch. 6
—	**Midterm Examination**	—
8	Recursion	Ch. 5
9	Trees & Binary Trees	Ch. 8
10	Graphs	Ch. 14
11	Sorting Algorithms (Merge Sort, Quick Sort, Heap Sort)	Ch. 12
12	Searching Algorithms (Linear, Binary)	Ch. 12
13	Hashing (Hash Tables, Collision Resolution)	Ch. 10
14	Review & Problem-Solving Practice	—
—	**Final Examination**	—

Note: The schedule above is subject to minor adjustments to accommodate the academic calendar and class progress.

7. Assessment and Grading

Assessment Breakdown

- Midterm Examination 30%
- Final Examination 50%
- Assignments 20%

Grading Scale

81–100	A	51–60	C
76–80	B+	46–50	D+
71–75	B	41–45	D
61–70	C+	0–40	F

8. Course Materials

- Lecture slides: ruslylove.github.io/datastruct
- PDF export of slides: slides.pdf
- Source repository: github.com/ruslylove/datastruct

This syllabus is subject to minor adjustments at the instructor's discretion to accommodate the academic calendar and class progress. Students are responsible for staying informed of any changes announced in class or via the course website.